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GROUP REPORT

Fifth Generation Fixed Network (F5G); F5G-A Methodology for Hybrid Measurement-Simulation of FTTR

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Foreword

This Group Report (GR) has been produced by ETSI Industry Specification Group (ISG) Fifth Generation Fixed Network (F5G).

Modal verbs terminology

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Executive summary

The present document presents a hybrid measurement-simulation methodology for the performance evaluation of Fiber-to-the-Room (FTTR) wireless devices in residential indoor environments. The methodology combines physical electromagnetic field measurements with numerical simulation techniques in a digital twin representation of the deployment environment.

FTTR systems combine optical access with multiple distributed wireless access points, creating complex indoor propagation conditions that are difficult to characterize using conventional testing approaches alone.

Conventional wireless testing approaches include fully physical measurement setups and purely numerical simulation models. While physical measurements provide traceability and realism, they are often constrained by test environment limitations and measurement effort. Simulation-based approaches offer flexibility and scalability, but their accuracy strongly depends on modelling assumptions. The hybrid methodology introduced in the present document aims to combine the strengths of both approaches by integrating measured field data into electromagnetic simulation workflows. The methodology is designed to facilitate realistic indoor performance evaluation while maintaining traceability and technical repeatability across different test environments.

The proposed framework enables the evaluation of key radio performance indicators such as received signal strength, power density distribution, and spatial coverage under realistic indoor conditions. By leveraging near-field measurement techniques and equivalent source modelling, the methodology allows the extension of physical measurements into representative digital twin environments, improving repeatability and reducing overall test complexity and measurement time.

The present document defines the reference model, data integration flow, validation principles, and a representative test case structure for hybrid measurement-simulation based FTTR performance evaluation. It is intended to support test laboratories, fixed network operators and service providers, and equipment designers in the development and application of consistent and reproducible hybrid testing methodologies.

1 Scope

The present document describes a hybrid measurement-simulation methodology for evaluating the wireless performance of Fiber-to-the-Room (FTTR) devices in residential indoor environments. The methodology applies to Sub FTTR Units (SFU) and Main FTTR Units (MFU) operating as access points within FTTR deployments.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] CTIA Certification: "Test Plan for Wireless Device Over-the-Air Performance, Multi-Probe Anechoic Chamber Methodology (CTIA 01.40)", Version 6.0.1.

[i.2] ETSI TR 137 977 (V19.0.0): "Universal Mobile Telecommunications System (UMTS); LTE; Universal Terrestrial Radio Access (UTRA) and Evolved Universal Terrestrial Radio Access (E-UTRA); Verification of radiated multi-antenna reception performance of User Equipment (UE) (3GPP TR 37.977 version 19.0.0 Release 19)".

[i.3] IEEE Std 1720™: "IEEE Recommended Practice for Near-Field Antenna Measurements".

[i.4] IEC 61000-4-21:2011: "Electromagnetic compatibility (EMC) - Part 4-21: Testing and measurement techniques - Reverberation chamber test methods".

[i.5] S. Gregson, J. McCormick, C. Parini: "Principles of Planar Near-Field Antenna Measurements", IET, 2023.

[i.6] L.J. Foged et al.: "Antenna measurement systems using multi-probe technology", 2015 IEEE™ Conference on Antenna Measurements & Applications (CAMA).

[i.7] S.-H. Park, G.-H. Jang, Y.-H. Seo, H.-S. Keum and S.-I. Bang: "High-Speed Antenna Measurement System Using Multi-Probe Array Technique for 5G Applications", Electronics, Vol. 11, No. 21, 2022, Article 3435.

[i.8] K. Persson and M. Gustafsson: "Reconstruction of equivalent currents using near-field data transformation with radome applications", Progress In Electromagnetics Research, Vol. 54, 2005, pp. 179–198.

[i.9] R. Bagheriasl et al.: "Toward Standardization of Phasor Near-field OTA Testing for Connected Devices – How RF Phasor Arrays Enable Augmented Measurements", IEEE™ Conference on Antenna Measurements & Applications (CAMA), 2025.

[i.10] Broadband Forum TR-398 Issue 3 Corrigendum 1: "Wi-Fi Residential & SOHO Performance Testing", May 2025.

[i.11] IEEE Std 145™-2025: "IEEE Standard for Definitions of Terms for Antennas".

[i.12] ETSI TS 138 161 (V19.3.0): "5G; NR; User Equipment (UE) TRP (Total Radiated Power) and TRS (Total Radiated Sensitivity) requirements; Range 1 Standalone and Range 1 Interworking operation with other radios (3GPP TS 38.161 version 19.3.0 Release 19) ".

[i.13] IEEE Std 802.11ax™-2021: "IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks--Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 1: Enhancements for High-Efficiency WLAN".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

digital twin: numerical model of a physical environment used to emulate realistic indoor propagation conditions and enable repeatable simulation of electromagnetic fields in that environment

equivalent electromagnetic source model: numerical representation derived from measured near-field data that reproduces the radiated field behaviour of the device under test in the region outside the equivalent-source region

hybrid measurement: combination of physical field measurement and electromagnetic simulation, where measured data are coupled with numerical computation to determine electromagnetic field distributions and performance metrics across defined test environments

measurement surface: geometrical surface associated with the spatial locations at which complex near-field electromagnetic field samples are acquired

near-field: region in the vicinity of an antenna corresponding to near-field conditions

NOTE: See IEEE Std 145-2025 [i.11] for antenna terminology.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

A_e	effective aperture of the receiving antenna (m^2)
A_{iso}	effective aperture of an isotropic antenna (m^2)
C_T	percentage of evaluation locations with received signal strength greater than or equal to T (%)
$C_{covered}$	percentage of evaluation locations classified as acceptable or strong (%)
C_{strong}	percentage of evaluation locations classified as strong (%)
$C_{acceptable}$	percentage of evaluation locations classified as acceptable (%)
C_{weak}	percentage of evaluation locations classified as weak (%)
C_{req}	minimum required percentage of evaluation locations classified as acceptable or strong for the selected test scenario (%)
G	receiving-antenna gain in linear units, including radiation efficiency and excluding impedance mismatch at the receiver interface
G_{dip}	linear gain of the lossless half-wave dipole receiving antenna
G_{ant}	linear gain of alternative receiving antenna
$I(.)$	indicator function, equal to 1 when its argument is true and 0 otherwise
η_{pol}	polarization mismatch factor between the incident electromagnetic field and the receiving antenna, in linear units, with $0 \leq \eta_{pol} \leq 1$
$\eta_{pol,dip}$	polarization mismatch factor for the lossless half-wave dipole reference antenna
$\eta_{pol,ant}$	polarization mismatch factor for the alternative receiving antenna
η_m	impedance mismatch efficiency between the receiving antenna and the receiver input, in linear units, with $0 \leq \eta_m \leq 1$
$\eta_{m,\,\mathrm{dip}}$	impedance mismatch efficiency of the half-wave dipole reference receiving antenna
$\eta_{m,\,\mathrm{ant}}$	impedance mismatch efficiency of the alternative receiving antenna

N	number of evaluation locations
PD	scalar power density, corresponding to the magnitude of the time-averaged Poynting vector (W/m²)
P_{av}	available received power from the receiving antenna, defined as the maximum power that can be delivered at the antenna terminals to a conjugate-matched receiver input (W)
P_r	actual received power delivered to the receiver input at the defined receiver reference plane (W)
$P_{r,i}$	actual received power delivered to the receiver input at evaluation location i (W)
$\overline{P_r}$	average power delivered to the receiver input, computed across all evaluation locations (W)
RSS_i	received signal strength at evaluation location i (dBm)
RSS_{avg}	average received signal strength (dBm)
T	received signal strength threshold used for spatial coverage evaluation (dBm)
T_{min}	minimum acceptable RSS boundary for the selected receiving antenna (dBm)
T_{strong}	strong received signal strength boundary for the selected receiving antenna (dBm)
T_{avg}	minimum required linear-power-averaged received signal strength for the selected test scenario and receiving antenna (dBm)
$T_{min,dip}$	minimum acceptable RSS boundary for the lossless half-wave dipole reference antenna (dBm)
$T_{strong,dip}$	strong RSS boundary for the lossless half-wave dipole reference antenna (dBm)
$T_{avg,dip}$	average received signal strength threshold for the half-wave dipole reference receiving antenna (dBm)
$T_{min,ant}$	minimum acceptable RSS boundary adapted to the alternative receiving antenna (dBm)
$T_{strong,ant}$	strong RSS boundary adapted to the alternative receiving antenna (dBm)
$T_{avg,ant}$	average received signal strength threshold adapted to the alternative receiving antenna (dBm)
λ	wavelength (m)

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

LAN	Local Area Network
MCS	Modulation and Coding Scheme
MFU	Main FTTR Unit
OTA	Over-The-Air
PD	Power Density
RF	Radio Frequency
RSS	Received Signal Strength
RSSI	Received Signal Strength Indicator
SFU	Sub FTTR Unit
TRP	Total Radiated Power
UDP	User Datagram Protocol
UE	User Equipment
WLAN	Wireless Local Area Network

4 Reference Model and Test Environment

4.1 Overview

Clause 4 defines the reference model used for the hybrid measurement-simulation methodology. The reference model describes the roles of the measurement and simulation subsystems and the data integration flow between them.

The framework enables the transformation of measured electromagnetic field data into equivalent numerical excitation sources for use within a digital twin representation of the indoor environment. This approach allows physically measured device behaviour to be propagated into representative deployment scenarios while maintaining consistency with electromagnetic theory.

The reference model is intended to remain implementation-independent and technology-neutral. It does not prescribe specific measurement hardware, numerical solvers, or commercial tools, but instead defines the functional blocks and data integration flow needed to support reproducible hybrid testing. The reference model does not define an overall system architecture for implementing the hybrid measurement-simulation methodology; such an architecture is for further study.

4.2 Measurement Subsystem

The measurement subsystem is responsible for acquiring electromagnetic field data from the device under test under controlled transmission conditions. Its primary objective is to capture physically representative field information that can be used as input for subsequent numerical simulation stages.

Near-field measurements are used to sample the complex electromagnetic field (amplitude and phase) in the vicinity of the device under test. The availability of phase-resolved field data is essential for the accurate computation of equivalent electromagnetic sources. These measurements are performed on one or more surfaces around the device, such as planar, cylindrical, spherical, cuboid or other suitable measurement geometries. The combined measurement geometry provides sufficient spatial coverage and sampling density to support the equivalent source computation.

During data acquisition, the device under test is operated under continuous transmission conditions using controlled wireless traffic generation. This ensures stable radiated emission behaviour throughout the measurement process and minimizes time-domain variations of the transmitted signals.

The measurement subsystem supports acquisition of complex electromagnetic field data, including both amplitude and phase, in order to enable equivalent source calculation and subsequent numerical simulation. This capability is achieved using measurement systems capable of coherent field sampling. Measurement systems using multiple probes enable parallel acquisition of field samples and can significantly reduce overall measurement time compared with sequential single-probe scanning approaches (see [i.6] and [i.7]).

EXAMPLE: Near-field phasor probe arrays are an example of measurement systems capable of coherent field sampling.

The present document does not prescribe specific probe designs, array geometries, scanner architectures, or mechanical positioning solutions. These implementation aspects may vary depending on laboratory infrastructure, provided that the measurement system satisfies the functional need of coherent complex-field acquisition with sufficient spatial sampling density.

4.3 Simulation Subsystem (Digital Twin)

The simulation subsystem provides the numerical representation of the environment used in the hybrid measurement-simulation methodology. It consists of a digital twin representing the indoor environment, combined with an electromagnetic solver capable of computing field propagation and spatial performance metrics.

The digital twin environment models the geometric layout and material characteristics of representative residential indoor deployment scenarios. The digital twin may include walls, floors, ceilings, and major furniture structures that influence radio wave propagation. The level of geometric and material detail may be adapted according to the intended accuracy and computational complexity of the simulation.

Equivalent electromagnetic sources derived from the measurement subsystem are integrated into the digital twin environment, either directly or through a compatible radiation representation derived from them, to reproduce the radiated behaviour of the device under test.

The simulation subsystem enables the computation of spatial field distributions and performance indicators, such as power density and received signal strength, at predefined locations within the digital twin environment.

4.4 Data Integration and Validation Flow

The data integration flow defines the process by which measured electromagnetic field data are transformed into numerical excitation models suitable for simulation within the digital twin environment. This transformation is performed using equivalent source modelling techniques that construct a source representation capable of reproducing the measured fields outside the measurement surface.

The data integration flow consists of the following stages:

- Acquisition of near-field measurement data around the device under test.
- Computation of equivalent electromagnetic source models based on the measured data.
- Integration of the equivalent sources into the numerical simulation environment, either directly or through a compatible radiation representation derived from them.
- Computation of spatial electromagnetic field distributions and extraction of performance metrics.

The equivalent source calculation is validated using reference antennas with known electromagnetic characteristics [i.9] by comparing field quantities obtained from the constructed equivalent sources with reference values for those antennas. The digital-twin propagation model may be validated separately by comparing simulated and measured field quantities at corresponding locations in the physical and digital environments. This staged validation provides additional diagnostic information and helps identify the main source of any discrepancy observed in the overall end-to-end comparison (see Annex C for an example).

4.5 Uncertainty and Repeatability Considerations

The accuracy and repeatability of hybrid measurement-simulation results depend on multiple factors spanning both the measurement and simulation domains. Measurement-related uncertainty contributors include measurement probe calibration, positioning accuracy, measurement system repeatability, and environmental stability during acquisition. Digital twin uncertainty contributors include differences between the physical environment and its geometric representation, uncertainty in the electromagnetic properties assigned to materials, transmitter and receiver placement and propagation model configuration and numerical settings, such as discretization parameters.

These contributions can act simultaneously, and their individual effects cannot in general be determined from the final RSS difference alone. A quantitative combined uncertainty requires separate characterization of the relevant contributors and their interactions. In the present methodology, the end-to-end comparison between simulated and measured RSS at corresponding receiver positions provides a practical assessment of their aggregate uncertainty effect.

The hybrid measurement-simulation methodology aims to support reproducible performance evaluation by defining consistent reference models, validation procedures, and data processing workflows. The development of a quantitative uncertainty budget, including the identification and characterization of dominant error sources, is foreseen as part of future refinements of the methodology and is for further study.

Laboratories applying the hybrid approach need to document relevant configuration parameters and environmental conditions to support result traceability and inter-laboratory comparison.

5 Test Case

5.1 Test Purpose

Clause 5 defines a hybrid measurement-simulation test case for the performance evaluation of FTTR devices in residential indoor environments. The test case is intended to demonstrate the application of the proposed methodology and to provide a reference workflow that can be reproduced. Extensions are for further study.

The purpose of the test case is to evaluate the wireless performance of Sub FTTR Units (SFU) and Main FTTR Units (MFU) under the indoor deployment conditions using the hybrid measurement-simulation methodology.

The test aims to characterize the spatial distribution of electromagnetic field quantities and radio performance indicators, such as received signal strength and power density, within a digital twin representation of the residential environment. The results are used to assess indoor coverage behaviour and to verify the consistency of the hybrid testing workflow.

5.2 Test Configuration

The test configuration consists of a device under test, which may be an SFU or an MFU, a wireless terminal, a traffic generator, and the measurement subsystem. The device under test is configured to operate under continuous transmission conditions using controlled wireless traffic in order to ensure stable radiated emission during measurement.

During the measurement phase, the device under test is positioned in a controlled near-field measurement setup, with measurements performed on multiple surfaces around the device. Figure 1 illustrates an implementation using six surfaces defining a rectangular cuboid. Near-field phasor probe arrays are used to acquire complex electromagnetic field samples on these surfaces.

The wireless terminal is positioned at a distance of 1 m from the device under test and remains connected throughout the measurement process. The traffic generator is connected on the LAN or uplink side and is used to generate continuous traffic.

5.3 Test Procedure

5.3.1 Phases

The hybrid performance evaluation test procedure is divided into three main phases:

1) Measurement phase;

2) Equivalent source calculation phase; and

3) Simulation-based performance evaluation phase.

The detailed workflow is described below.

5.3.2 Measurement Phase

1) One WLAN terminal is connected to the SFU/MFU under test and positioned at a distance of 1 m. The WLAN terminal remains connected during the entire test.

2) Unless otherwise described in this procedure, the WLAN parameters (mode, channel, authentication and encryption method, etc.) of the MFU and SFU use the manufacturer default settings. The channel bandwidth is set to 20 MHz for both 2,4 GHz and 5 GHz. Other channel bandwidths supported by the MFU/SFU may be evaluated using the same procedure. For the test case described here, the number of spatial streams is set to two (Nss = 2), where supported by the device under test, using IEEE 802.11ax [i.13]. If two spatial streams are not supported, a supported spatial-stream configuration is used and documented. Other supported spatial-stream configurations or WLAN protocols may be evaluated using the same procedure. Any alternative WLAN configuration is evaluated as a separate test run. The WLAN configuration used for each test run is documented to support comparability and repeatability.

3) Generate continuous UDP traffic between the device under test and the WLAN terminal so that the device under test remains actively transmitting during the measurement.

NOTE: UDP is used in the reference test setup for ease of continuous traffic generation and monitoring. The UDP traffic may be carried over Ethernet on the wired side of the setup. Other means of generating a controlled continuous traffic stream may be used, provided that the device under test remains actively transmitting throughout the measurement.

4) After initiating data transmission, wait at least 120 seconds as a stabilization period to allow the wireless link and transmission to reach steady-state operation before measurement begins.

5) Maintain the data transmission continuously throughout the entire data acquisition process.

6) Measure near-field phasor data (both phase and amplitude) on multiple surfaces around the SFU/MFU.

EXAMPLE: Measurements can be performed on six surfaces defining a rectangular cuboid around the SFU/MFU as shown in figure 1.

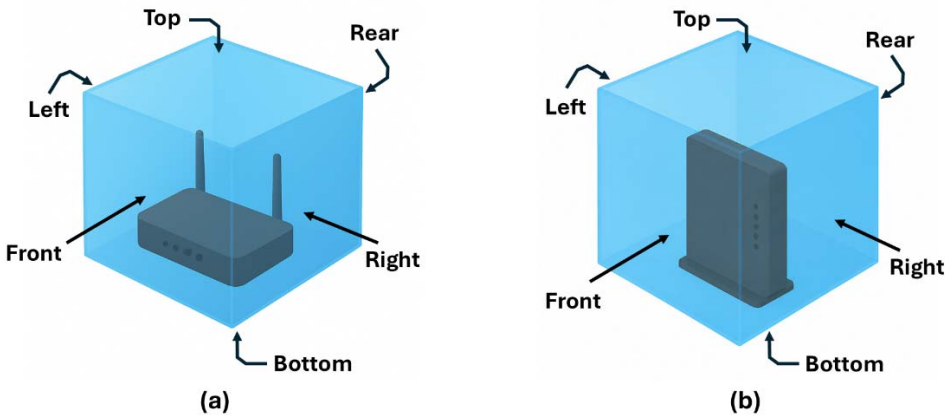


Figure 1: Example rectangular cuboid surrounding SFU/MFU: (a) with external antennas; (b) with internal antennas

7) During step 6 of the measurement phase, continuously monitor the throughput at the terminal to ensure that the device under test remains actively transmitting; values near zero indicate that the intended traffic condition is not maintained. If this occurs, re-establish the traffic condition starting from Step 3 and the stabilization period is repeated before the affected data are acquired again.

5.3.3 Equivalent Source Calculation Phase

1) Process the measured near-field data to compute an equivalent electromagnetic source representation of the device under test. The equivalent source model is selected to enable accurate reproduction of the measured field distribution outside the measurement surfaces and compatibility with the selected numerical simulation environment.

5.3.4 Simulation-Based Performance Evaluation Phase

1) Integrate the equivalent sources derived from clause 5.3.3 into the digital twin environment, either directly or through a compatible radiation representation derived from them.

2) Perform numerical electromagnetic simulations to compute power density and received signal strength, at evaluation locations defined according to clause B.5.

3) Generate and report spatial performance maps and coverage indicators. These results form the basis for the performance evaluation of the SFU or MFU under test.

5.4 Expected Result

5.4.1 Overview

The following physical quantities are considered:

- Power Density (PD): magnitude of the time-averaged Poynting vector at each evaluation location.
- Received Signal Strength (RSS): actual power delivered to the receiver input at each evaluation location, expressed in dBm and evaluated at a documented receiver reference plane using a receiving antenna model with documented gain, polarization, orientation, and impedance-matching assumptions.

5.4.2 Performance Criteria

Performance evaluation is based on two complementary characteristics:

- the spatial distribution of received signal strength across the predefined evaluation locations; and
- the linear-power-averaged received signal strength across the same evaluation locations.

For a selected received signal strength threshold T, the corresponding spatial coverage percentage is calculated as:

$$C_T = \frac{100}{N}\sum_{i=1}^{N} I(RSS_i \geq T) \quad (1)$$

where RSS_i is the received signal strength at evaluation location i, $N$is the number of evaluation locations, and $I(\cdot)$ is the indicator function, which is equal to 1 when the stated condition is satisfied and 0 otherwise.

To characterize the spatial received signal strength distribution, two threshold boundaries are selected before the test: a minimum acceptable boundary $T_{\min}$ and a strong-signal boundary T_{strong}, where:

$$T_{\min} < T_{\text{strong}}$$

Each evaluation location is classified according to the following ranges:

- Strong: $RSS_i \geq T_{\text{strong}}$
- Acceptable: $T_{\min} \leq RSS_i < T_{\text{strong}}$
- Weak: $RSS_i < T_{\min}$

The corresponding spatial percentages are defined as follows:

- $C_{\text{covered}} = C_{T_{min}}$, representing the percentage of locations classified as acceptable or strong;
- $C_{\text{strong}} = C_{T_{\text{strong}}}$ representing the percentage of locations classified as strong;
- $C_{\text{acceptable}} = C_{\text{covered}} - C_{\text{strong}}$, representing the percentage of locations classified as acceptable;
- $C_{\text{weak}} = 100\ \% - C_{\text{covered}}$, representing the percentage of locations classified as weak.

The spatial coverage criterion is:

$$C_{\text{covered}} \geq C_{\text{req}} \quad (2)$$

where C_{req} is the minimum needed percentage of covered evaluation locations classified as acceptable or strong for the selected test scenario.

The linear-power-averaged received signal strength is calculated by first converting the received signal strength at each evaluation location from dBm to linear received power:

$$P_{r,i}[W] = 10^{\frac{RSS_i[dBm]-30}{10}} \quad (3)$$

where $P_{r,i}$is the actual power delivered to the receiver input at the defined receiver reference plane for evaluation location i. The linear received powers are then averaged:

$$\bar{P_r} = \frac{1}{N}\sum_{i=1}^{N} P_{r,i} \quad (4)$$

and the resulting average received power is converted back to dBm:

$$RSS_{avg}[dBm] = 10\log_{10}(\bar{P_r}[W]) + 30 \quad (5)$$

The quantities RSS_i, $P_{r,i}$, $\bar{P_r}$, and RSS_{avg} refer to power delivered to the receiver input at the defined receiver reference plane. Receiving-antenna gain, polarization mismatch, and impedance mismatch are therefore already reflected in the calculated RSS values. These factors are not applied again during the averaging process. The corresponding aggregate mean-power criterion is:

$$RSS_{\text{avg}} \geq T_{\text{avg}} \quad (6)$$

where T_{avg} is the minimum needed linear-power-averaged received signal strength for the selected test scenario.

The device configuration satisfies the selected performance criteria only if both the spatial coverage criterion and the mean-power criterion are satisfied, hence:

$$C_{\text{covered}} \geq C_{\text{req}} \quad \text{and} \quad RSS_{\text{avg}} \geq T_{\text{avg}} \quad (7)$$

C_{covered} limits the proportion of weak-coverage locations, while RSS_{avg} characterizes the mean received power across the complete evaluation set. The two criteria provide complementary information and are evaluated separately. Both are calculated from the same spatial RSS distribution and are therefore related, but neither criterion is derived from the other. A high linear-power-averaged RSS does not necessarily imply good spatial coverage, since a limited number of high-RSS locations can increase the average while other locations remain below the minimum acceptable boundary. Conversely, a high covered percentage does not necessarily imply a high average RSS if most covered locations are close to the minimum acceptable boundary.

The applicable values of T_{min}, T_{strong}, T_{avg}, and C_{req} are determined in accordance with clause 5.4.3 and documented before evaluating the device under test. The same values are applied consistently to all devices and configurations being compared and are not adjusted according to the obtained results.

The reported results include spatial coverage maps together with C_{strong}, $C_{\mathrm{acceptable}}$, C_{weak}, C_{covered}, and RSS_{avg}. The receiving-antenna gain, polarization, and impedance-matching assumptions are already reflected in the calculated RSS_ivalues and in the threshold adaptation described in clause 5.4.3. No additional gain, polarization mismatch, or impedance mismatch factor is applied in equations (1) to (5).

5.4.3 Reference antenna and threshold-range adaptation

The RSS levels applicable to an indoor coverage evaluation depend on the service objective, propagation environment, transmitter power, receiving-device characteristics, and interference conditions.

Broadband Forum TR-398 [i.10] uses approximate target RSSI values of -45 dBm, -60 dBm, and -75 dBm for close, medium, and far test conditions, respectively, under defined laboratory configurations. These values illustrate that the relevant RSS level depends on the specified propagation and test conditions; they are not applied directly as the performance boundaries in the present document.

For the reference residential environment evaluation scenario defined in the present document, the following RSS boundaries are used for a lossless half-wave dipole reference receiving antenna oriented for polarization matching and conjugate-matched to the receiver input. The corresponding reference parameters are $G_{\mathrm{dip}} = 1{,}64$, $\eta_{\mathrm{pol,\,dip}} = 1$ and $\eta_{\mathrm{m,\,dip}} = 1$.

where G_{dip} is the linear gain of the lossless half-wave dipole reference receiving antenna, $\eta_{\mathrm{pol,dip}}$ is its polarization mismatch factor, and $\eta_{\mathrm{m,dip}}$ is the impedance mismatch efficiency between the reference antenna and the receiver input. The values $\eta_{\mathrm{pol,dip}} = 1$ and $\eta_{\mathrm{m,dip}} = 1$ represent perfect polarization matching and conjugate matching, respectively.

The RSS boundaries are $T_{min,dip} = -70$ dBm and $T_{strong,dip} = -50$ dBm.

These boundaries define the following spatial RSS ranges:

- **Strong:** $RSS_i \geq T_{\mathrm{strong,dip}}$, corresponding to $RSS_i \geq -50$ dBm;
- **Acceptable:** $T_{\mathrm{min,dip}} \leq RSS_i < T_{\mathrm{strong,dip}}$, corresponding to -70 dBm $\leq RSS_i < -50$ dBm;
- **Weak:** $RSS_i < T_{\mathrm{min,dip}}$, corresponding to $RSS_i < -70$ dBm.

The reference values used in the present document are not derived directly from the Broadband Forum TR-398 [i.10] close, medium and far target values. For the reference residential evaluation scenario, -70 dBm is selected as the minimum acceptable boundary and -50 dBm as the strong-signal boundary, defining a 20 dB classification interval with -60 dBm at its midpoint on the dB scale. These values are engineering reference values for the present methodology. Different boundaries can be specified before evaluation where required by a different service objective, deployment scenario or receiving-device requirement.

For evaluations using the reference half-wave dipole receiving antenna, the reference RSS boundaries defined above are used. When an alternative receiving antenna or probe is used, the corresponding adapted boundaries obtained using equation (8) are used. For the reference residential evaluation scenario, the minimum percentage of locations classified as acceptable or strong is $C_{\mathrm{req}} = 90\ \%$.

The value of 90 % is selected as a reference coverage objective for the residential evaluation scenario, allowing a limited proportion of weak locations while requiring acceptable-or-strong coverage over most of the evaluation set. This reference value is intended for a spatially distributed set of evaluation locations representative of the residential environment, including user positions relevant to the test scenario and coverage-critical areas. The applicable value also depends on the selection of evaluation locations. For example, a higher required percentage may be appropriate when the evaluation set contains only critical user locations, whereas a lower value may be appropriate when the set provides broader spatial sampling that also includes less critical locations. The evaluation-location selection method and the corresponding value of C_{req} are defined before the performance results are analysed.

For the reference residential evaluation scenario, the threshold for the linear-power-averaged received signal strength is $T_{\text{avg, dip}} = -60$ dBm.

This threshold represents the minimum required mean received power for the half-wave-dipole reference receiving antenna. For the reference residential evaluation scenario, T_{avg} = -60 dBm is selected as an intermediate reference level between the strong-signal boundary T_{strong} = -50 dBm and the minimum acceptable boundary T_{min} = -70 dBm. It provides a complementary mean-power criterion intended to prevent satisfactory spatial coverage from being obtained mainly from locations close to the minimum acceptable boundary. The applicable value of T_{avg} can be adjusted for a different service objective or deployment scenario, taking into account factors such as the expected service level, transmitter configuration, propagation environment and receiving-device requirements. The selected value is defined before the performance results are analysed and is applied consistently to the devices or configurations being compared.

This mean-power criterion is applied together with the spatial coverage criterion defined in clause 5.4.2. Consequently, for the half-wave-dipole reference receiving antenna, the reference residential evaluation criteria are:

- $C_{\text{covered}} \geq 90$ % and $RSS_{\text{avg}} \geq -60$ dBm

When an alternative receiving antenna or probe is used, the spatial RSS boundaries and the average received signal strength threshold are adapted according to the receiving-antenna gain, the polarization mismatch between the incident electromagnetic field and the receiving antenna, and the impedance mismatch between the receiving antenna and the receiver input:

$$T_{k,\text{ant}} = T_{k,\text{dip}} + 10log_{10}\left(\frac{G_{\text{ant}}\eta_{\text{pol,ant}}\eta_{\text{m,ant}}}{G_{\text{dip}}\eta_{\text{pol,dip}}\eta_{\text{m,dip}}}\right), \qquad k \in \{\text{min, strong, avg}\} \tag{8}$$

Where:

- G_{ant} is the linear gain of the alternative receiving antenna, excluding receiver-interface impedance mismatch;
- G_{dip} is the linear gain of the lossless half-wave dipole reference receiving antenna;
- $\eta_{\text{pol,ant}}$ and $\eta_{\text{pol, dip}}$ are the corresponding polarization mismatch factors;
- $\eta_{\text{m, ant}}$ and $\eta_{\text{m, dip}}$ are the corresponding impedance mismatch efficiencies between each receiving antenna and its receiver input.

The antenna terminology used in this clause is consistent with [i.11].

Equation (8) is applied separately to:

- $T_{\text{min,dip}}$ to obtain $T_{\text{min,ant}}$;
- $T_{\text{strong,dip}}$ to obtain $T_{\text{strong,ant}}$; and
- $T_{\text{avg,dip}}$ to obtain $T_{\text{avg,ant}}$;

Applying the same gain, polarization, and impedance-mismatch correction to $T_{\text{min,dip}}$ and $T_{\text{strong,dip}}$ preserves the width of the three spatial RSS ranges.

If the reference and alternative receiving antennas are both perfectly polarization matched to the incident field and conjugate-matched to their respective receiver inputs, then:

- $\eta_{\text{pol,ant}} = \eta_{\text{pol,dip}} = 1$

and

- $\eta_{\text{m, ant}} = \eta_{\text{m, dip}} = 1$

Under these assumptions, equation (8) reduces to gain-only threshold adaptation.

The adaptation in equation (8) applies when the receiving-antenna gain, polarization mismatch and impedance mismatch can be represented by common values across the evaluation locations. Where the receiving-antenna response varies significantly with arrival direction, incident polarization, evaluation location or operating frequency, a single common set of adapted thresholds $T_{\min}$, T_{strong} and T_{avg} is not sufficient. Such cases need location-dependent receiver modelling and threshold treatment and are outside the reference adaptation procedure defined in this clause.

The receiving-antenna gain, polarization, orientation, impedance-matching condition, receiver reference plane, and applicable mismatch assumptions are documented as part of the test configuration. The thresholds used for evaluation correspond to the same receiving-antenna and reference-plane assumptions as the calculated RSS values.

When a different deployment scenario or service objective is evaluated, any alternative threshold values are specified in the test configuration before the device results are analysed.

5.4.4 Physical relations

Power density and received signal strength are physically related through the characteristics of the receiving antenna and the power transfer between the antenna and the receiver input. The relations in this clause are expressed for a single evaluation location. Aggregation across evaluation locations is performed according to clause 5.4.2.

In this clause, PD denotes the scalar power density, corresponding to the magnitude of the time-averaged Poynting vector. The polarization mismatch factor η_{pol} represents the polarization coupling between the incident electromagnetic field and the receiving antenna, with:

- $0 \leq \eta_{\text{pol}} \leq 1$

The value of $\eta_{\text{pol}} = 1$ corresponds to perfect polarization matching.

The available received power P_{av} is the maximum power that the receiving antenna can deliver to a conjugate-matched receiver input. Conjugate matching is obtained when the receiver input impedance equals the complex conjugate of the receiving-antenna impedance. The available received power is:

$$P_{\text{av}} = PD \cdot A_e \cdot \eta_{\text{pol}} \tag{9}$$

The impedance mismatch efficiency η_{m} represents the fraction of the available received power that is delivered to the receiver input:

$$\eta_m = \frac{P_r}{P_{\text{av}}}, \qquad 0 \leq \eta_{\text{m}} \leq 1. \tag{10}$$

The actual power delivered to the receiver input at the defined receiver reference plane is:

$$P_r = P_{\text{av}}\eta_m = PD \cdot A_e \cdot \eta_{\text{pol}} \cdot \eta_m \tag{11}$$

where P_{av} is the available received power from the receiving antenna (W), P_r is the actual power delivered to the receiver input at the defined reference plane (W), PD is the scalar power density (W/m²), A_e is the effective aperture of the receiving antenna (m²), η_{pol} is the polarization mismatch factor, and η_{m} is the impedance mismatch efficiency.

Under conjugate matching, $\eta_{\text{m}} = 1$ and $P_r = P_{\text{av}}$. Otherwise, $\eta_{\text{m}} < 1$ and $P_r < P_{\text{av}}$.

The effective aperture is expressed as the product of the antenna gain and the effective aperture of an isotropic antenna:

$$A_e = G \cdot A_{iso} \tag{12}$$

with:

$$A_{iso} = \frac{\lambda^2}{4\pi} \tag{13}$$

where G is the antenna gain in linear units, A_{iso} is the effective aperture of an isotropic antenna, and λ is the wavelength (m) given by:

$$\lambda = \frac{c}{f} \tag{14}$$

where f is the operating frequency (Hz) and c is the speed of light (m/s).

In the present document, G denotes antenna gain and therefore includes radiation efficiency but excludes impedance mismatch between the antenna and the receiver input. If realized gain is used instead, impedance mismatch is already included in the gain value and η_{m} is set to 1 to avoid double counting.

The received signal strength expressed in logarithmic units (dBm) is related to the received power in linear units (W) as follows:

$$\mathrm{RSS}\ [\mathrm{dBm}] = 10 \cdot \log_{10}(P_r\ [W]) + 30 \tag{15}$$

The inverse relation is:

$$P_r[W] = 10^{\frac{RSS[dBm]-30}{10}} \tag{16}$$

Polarization mismatch and impedance mismatch are included in the relationship between incident power density and delivered received power through equation (11). They are not included explicitly in equations (12) to (16), since the effective-aperture definition, wavelength relation, and mathematical conversion between watts and dBm are independent of these mismatch factors.

Combining the above relations allows conversion between received signal strength and power density. The corresponding expressions are:

$$PD\,[W/\mathrm{m}^2] = \frac{P_r}{A_e \eta_{\mathrm{pol}} \eta_m} \tag{17}$$

Or equivalently:

$$PD\,[W/\mathrm{m}^2] = \frac{1}{G A_{iso} \eta_{\mathrm{pol}} \eta_m} \cdot 10^{\frac{RSS[dBm]-30}{10}} \tag{18}$$

The received signal strength corresponding to a given power density is:

$$RSS\ [\mathrm{dBm}] = 10 \cdot \log_{10}\left(PD\ [W/\mathrm{m}^2] \cdot G A_{iso} \cdot \eta_{\mathrm{pol}} \cdot \eta_m\right) + 30 \tag{19}$$

These relations provide the physical basis for conversion between incident power density and received signal strength at the defined receiver reference plane, including receiving-antenna gain, polarization mismatch, and impedance mismatch. They are not applied separately in the performance criteria of clause 5.4.2, since the calculated RSS values and the adapted threshold values already incorporate the applicable receiving-antenna and receiver-interface assumptions.

Losses occurring after the defined receiver reference plane, such as cable, connector, or receiver-front-end losses, are not included in η_{m}unless they are explicitly incorporated into the receiver model. Any such losses are documented separately and applied consistently to both the calculated RSS values and the associated thresholds.

Annex A:
Comparison of Hybrid Measurement-Simulation Methodology with Conventional Test Approaches

A.1 Overview of FTTR Performance Evaluation Approaches

Wireless performance evaluation of FTTR systems can be performed using several different testing approaches, including shielded over-the-air testing, radiated chamber measurements, real home or field testing, and simulation-based evaluation. These approaches do not produce identical categories of results and are designed to address different evaluation objectives.

Some methods primarily characterize device-level radio parameters or antenna radiation behaviour under controlled propagation conditions, while others focus on empirical performance observation in real deployments or on modelled spatial propagation in virtual environments. As a result, the choice of test methodology directly influences the type of performance indicators that can be obtained, such as radiation patterns, integrated radiated power metrics, point-based signal measurements, or spatial coverage maps.

This annex provides a qualitative comparison of these approaches and positions the hybrid measurement–simulation methodology defined in the present document as a complementary framework that combines measured device radiation behaviour with environment-aware spatial performance evaluation.

A.2 Shielded and Radiated OTA Testing

Shielded Over-The-Air (OTA) testing refers to measurement configurations in which the device under test is placed inside electrically shielded enclosures or connected through controlled radiative or conductive coupling mechanisms. These setups are commonly used to isolate the device from external interference and to create stable and repeatable test conditions.

Conductive or semi-radiative shielded test setups can rely on RF cables, couplers, or small radiative enclosures. While these methods provide high repeatability and low sensitivity to external interference, they do not capture realistic antenna-environment interactions and do not represent residential indoor deployment geometries.

Radiated OTA testing performed in anechoic or reverberation chambers enables free-space or statistically isotropic propagation measurements. Anechoic chambers intentionally suppress reflections to emulate free-space radiation behaviour, while reverberation chambers generate randomized multipath conditions. Although these methods are suitable for antenna characterization and total radiated power measurements, they do not explicitly model residential apartment layouts, wall structures, or deterministic indoor propagation paths (see [i.1], [i.2], [i.3] and [i.4]). As a result, their ability to represent realistic FTTR deployment scenarios remains limited.

A.3 Real Home and Field Testing

Real home or field testing evaluates FTTR network in actual residential environments, providing high realism and direct observation of in-situ performance behaviour. Such testing naturally includes building materials, room layouts, furniture, and real user interference conditions.

Despite this realism, real home testing presents significant limitations in terms of measurement control and repeatability. Environmental conditions such as neighbouring wireless interference, human movement, furniture arrangement, and building-specific characteristics cannot be fully controlled or reproduced across different test campaigns or laboratories. These factors significantly limit the ability to perform systematic comparison, controlled validation, and long-term repeatable performance assessment (see [i.1]).

A.4 Simulation-Only Performance Evaluation

Simulation-based performance evaluation relies on numerical modelling of antennas and indoor propagation environments without direct physical measurement input. These approaches provide high flexibility, scalability, and rapid exploration of multiple deployment scenarios.

However, the accuracy of simulation-only methods strongly depends on the fidelity of antenna models, material parameter assumptions, and environmental abstractions. Simplified antenna representations and idealized indoor models may not fully capture the complex radiation behaviour of integrated FTTR devices. As a result, purely simulation-based approaches may lack physical traceability and may need extensive validation against measured data (see [i.2]).

A.5 Hybrid Measurement-Simulation Methodology

The hybrid measurement-simulation methodology defined in the present document combines near-field electromagnetic measurement with numerical propagation modelling in a digital twin environment. Measured complex field data are used to construct equivalent electromagnetic sources that represent the radiated behaviour of the device under test (see [i.5], [i.6], [i.8] and [i.9]). These sources are subsequently propagated within a modelled residential environment.

This approach preserves physical traceability through direct measurement while enabling flexible and scalable evaluation of realistic indoor deployment scenarios. By separating device radiation characterization from environment-specific propagation modelling, the hybrid methodology improves reproducibility compared to field testing and enhances realism compared to purely synthetic simulation models.

The hybrid framework provides a balanced compromise between measurement realism, repeatability, and computational flexibility, supporting systematic performance evaluation of FTTR devices under representative residential conditions.

A.6 Qualitative Comparison Summary

Table A.1 provides a qualitative comparison of the main FTTR performance evaluation approaches discussed in this annex. For the purpose of qualitative comparison, the following criteria are used. Measurement realism refers to the degree to which the test environment and propagation conditions represent realistic residential FTTR deployment scenarios. Environment flexibility refers to the ability to configure and evaluate different indoor layouts, material properties, and deployment scenarios without extensive physical reconfiguration. Reproducibility refers to the ability to obtain consistent and comparable results across repeated test campaigns and different laboratories. Measurement effort refers to the practical complexity, time consumption, and operational resources needed to perform the test procedure. Scalability refers to the capability to efficiently apply the methodology to a large number of devices, configurations, or test scenarios. Physical traceability refers to the extent to which the obtained results are directly grounded in measured physical electromagnetic behaviour of the device under test.

Table A.1: Qualitative comparison of FTTR performance evaluation approaches

Test Approach	Measurement Realism	Environment Flexibility	Reproducibility	Measurement Effort	Scalability	Physical Traceability
Shielded OTA (Conductive / Radiative)	Low to Medium	Low	High	Low	High	High
Radiated OTA Chamber Testing	Medium	Low	High	Medium	Medium	High
Real Home / Field Testing	High	High	Low	High	Low	Medium
Simulation-Only Evaluation	Low to Medium	High	High	Low	High	Low to Medium
Hybrid Measurement-Simulation (This Methodology)	High	High	High	Medium	High	High

The hybrid measurement-simulation methodology provides a balanced combination of physical realism, environment flexibility, reproducibility, and scalability. It complements existing test approaches by enabling realistic residential indoor performance evaluation while maintaining measurement traceability and controlled experimental conditions.

In contrast to conventional approaches discussed above, which focus on antenna radiation characteristics, device-level radio metrics, or empirical field measurements, the hybrid measurement-simulation methodology enables the generation of spatial performance indicators that combine physical measurement traceability with realistic indoor propagation modelling. This includes coverage maps, power density distributions, and received signal strength evaluations within representative residential deployment environments. As such, the hybrid approach extends the scope of performance evaluation beyond traditional single-domain test methods.

Annex B:
Reference Residential Digital Twin Environment Framework

B.1 Purpose and Scope

This annex defines a reference framework for residential indoor digital twin environments used in the hybrid measurement-simulation methodology. The objective is to enable reproducible and comparable simulation-based performance evaluation across different laboratories and toolchains while maintaining representative residential deployment characteristics.

The framework defines parametric geometry templates, material classes, and reference coordinate conventions that allow consistent residential FTTR deployment scenarios.

B.2 Reference Geometry Model

The reference digital twin environment is based on a simplified residential apartment geometry composed of multiple connected rooms. The geometry model includes at least the following functional areas:

- One primary living space where the main FTTR unit is located.
- One or more secondary rooms representing bedrooms or offices.
- Internal corridors or transitional spaces connecting the rooms.

The geometry is defined using parametric dimensions, including room length, width, height, wall thickness, and doorway openings. Parameter ranges may be selected to represent residential building configurations relevant to the test scenario while allowing flexibility to adapt the environment size to specific test scenarios.

The geometry model avoids excessive architectural detail and instead focus on major structural features that significantly affect radio propagation, such as wall placement, floor separation, and room connectivity.

B.3 Reference Material Classes

To ensure reproducibility without enforcing proprietary construction details, the digital twin framework defines a set of reference material classes rather than specific commercial building materials. Each material class is characterized by electromagnetic parameters, such as relative permittivity, conductivity, and loss tangent.

The following reference material classes are recommended:

- Lightweight internal wall (e.g. drywall or plasterboard equivalent)
- Structural wall (e.g. concrete or brick equivalent)
- Floor/ceiling slab
- Glass partition or window surface
- Wooden or composite furniture surfaces

For each material class, parameter ranges may be defined based on commonly reported values in the literature and building practice. Laboratories applying the methodology document the selected parameter values to support result traceability and comparison.

B.4 Coordinate System and Device Placement

A common reference coordinate system is defined for the digital twin environment. The origin of the coordinate system is placed at the geometric centre of the main FTTR unit.

The orientation of the coordinate axes is aligned with the principal walls of the reference geometry. Device placement height and orientation is documented relative to the floor reference plane and coordinate axes.

This coordinate convention enables consistent positioning of equivalent sources and evaluation locations across different simulation platforms.

B.5 Evaluation Location Grid

Performance evaluation locations within the digital twin environment are defined using a discrete set of spatial sampling points. These locations represent user positions relevant to the test scenario and coverage-critical areas in residential FTTR deployments.

Evaluation locations may be defined using one or more of the following approaches:

- Uniform three-dimensional grid sampling within rooms.
- Planar sampling at representative user device heights.
- Discrete points corresponding to usage positions relevant to test scenario such as desks, sofas, or bedroom areas as defined by the test scenario.

The three approaches differ primarily in the spatial sampling strategy used to evaluate performance within the digital twin environment.

Uniform three-dimensional grid sampling provides the most comprehensive representation, as it covers the full volume of the environment. This approach enables detailed spatial characterization of electromagnetic field distributions but results in many evaluation points, increasing data processing and analysis complexity. In addition, some sampled locations may not correspond to user positions relevant to the test scenario.

Planar sampling at representative user heights reduces the number of evaluation points and provides a more efficient evaluation while maintaining relevance to common usage scenarios. However, this approach does not capture variations across different heights within the environment.

Non-uniform discrete point sampling focuses on a limited set of representative locations corresponding to usage positions defined by the test scenario. This approach provides an efficient and scenario-relevant evaluation, while reducing the number of evaluation points. However, it provides less spatial coverage compared to grid-based approaches.

The number and spatial distribution of evaluation locations are selected to capture spatial field variations while maintaining reasonable data processing complexity. Representative evaluation heights corresponding to user device positions defined by the test scenario (e.g. handheld or desktop usage) are considered.

B.6 Reproducibility and Documentation Principles

To support reproducibility and inter-laboratory comparison, the digital twin environment configuration used for a given test campaign is documented. The present documentation includes geometry parameters, material class assignments, electromagnetic parameter values, coordinate definitions, and evaluation location coordinates.

By using parametric reference templates and standardized documentation practices, the hybrid measurement-simulation methodology enables consistent application across different laboratories and simulation toolchains while preserving flexibility for scenario adaptation.

Annex C:
Illustrative validation and application of the hybrid measurement-simulation methodology

C.1 Purpose

This annex applies the methodology to one SFU configuration at 5,18 GHz. It covers the free-space radiation comparison, validation of the indoor digital twin, and comparison of simulation-only and hybrid measurement-simulation results in the same indoor environment. The digital-twin validation was performed separately at 5,2 GHz using an omnidirectional reference antenna and indoor measurements at 44 receiver positions.

Annex C is regarded as an example for a particular setup, further scenarios are for further study.

C.2 Test and Simulation Configuration

The example considers an SFU operating at 5,18 GHz. The configuration used during the near-field measurement phase is summarized in Table C.1. The configuration used for indoor propagation calculation is summarized in Table C.2.

Table C.1: Near-field measurement configuration

Parameter	Configuration
Device under test	SFU
Operating frequency	5,18 GHz
WLAN channel	36
Channel bandwidth	20 MHz
MCS	5
Transmission configuration	Single-antenna transmission
Active antenna	5 GHz band antenna identified as "Antenna 0" in the test campaign

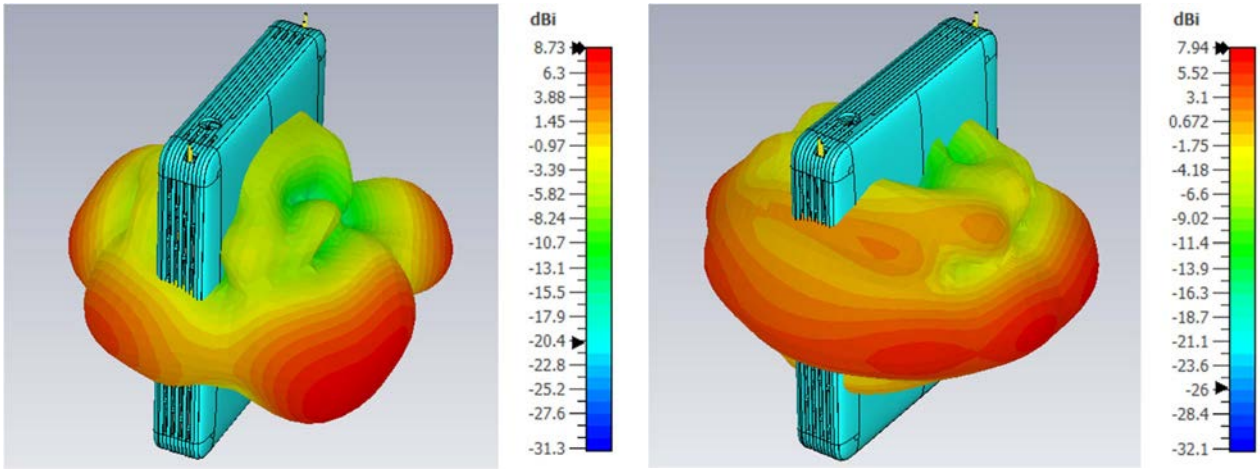
Table C.2: Indoor propagation configuration

Parameter	Configuration
Indoor environment	Reference residential digital twin
Propagation software and method	Sionna ray-tracing
Evaluation set	44 receiver positions
Receiver height	0,66 m
Configured transmitted power in Sionna	23 dBm

C.3 Free-space Radiation Comparison

The equivalent-source calculation approach used in this example relies on phasor near-field methods previously validated through comparisons with anechoic-chamber measurements and numerical simulations [i.9]. A full validation of the calculation methodology is therefore not repeated in this annex. For the specific SFU configuration considered here, the free-space radiation characteristics calculated from the equivalent sources derived from the near-field measurements are compared with the corresponding simulation-only results. The comparison includes the three-dimensional far-field radiation pattern, directivity and total radiated power.

The measured near-field data were processed to calculate the far-field pattern of the complete SFU. Figure C.1 compares this pattern with the full-device simulation result in CST for the same antenna and frequency. The hybrid-measurement far-field pattern result is also imported into CST for visualization.



(a) (b)

Figure C.1: Far-field gain patterns of SFU Antenna 0 at 5,18 GHz: (a) Measurement-derived pattern; (b) Simulation-derived pattern

The two far-field patterns show similar overall radiation behaviour, with the main radiation regions reproduced in both cases and differences in the detailed lobe shape. For the purpose of this example, a difference of 2 dB is selected as a reference comparison level for the reported directivity and TRP values. This value is used as an engineering comparison criterion rather than as a universal antenna-performance acceptance limit. For context, ETSI TS 138 161 [i.12] reports preliminary example 95 % expanded uncertainties of 1,73 dB and 1,96 dB for anechoic-chamber TRP measurements above 3 GHz, depending on the UE usage configuration. The 3GPP values relate specifically to TRP measurement uncertainty and do not define a directivity acceptance limit. As summarized in Table C.3, the directivity and TRP differences are 1,76 dB and 0,70 dB respectively. Both values are below the 2 dB reference comparison level used in this example.

Table C.3: Directivity and TRP Comparison at 5,18 GHz

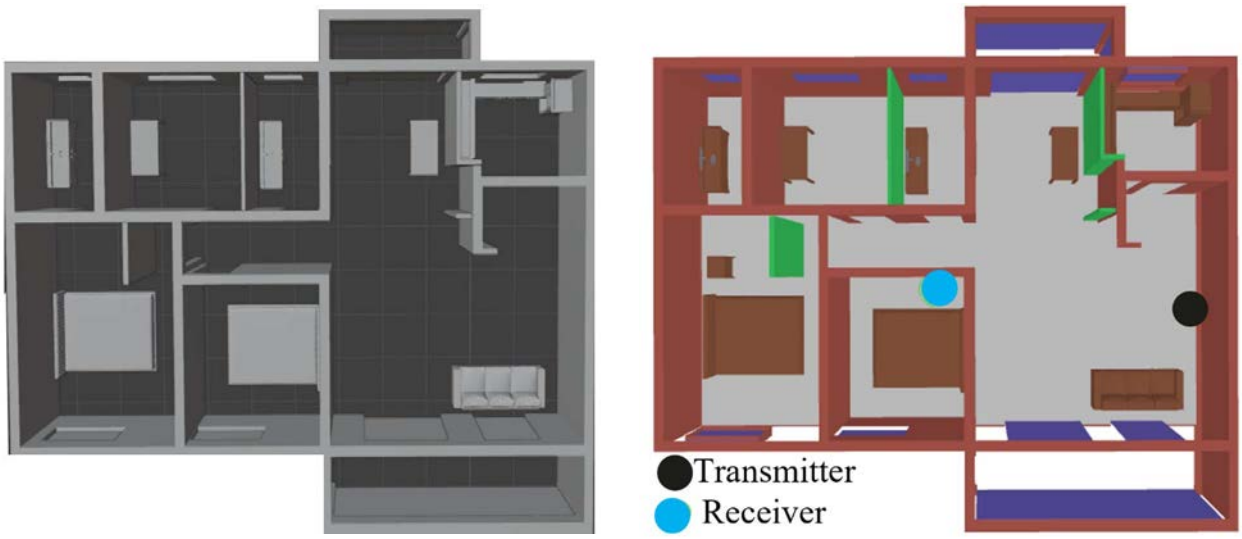
Quantity	Simulation	Measurement	Difference
Directivity	7,99 dBi	9,75 dBi	1,76 dB
TRP	22,50 dBm	23,20 dBm	0,70 dB

C.4 Validation of the Indoor Digital Twin

A real residential test apartment located in China, was selected as the reference indoor environment. A three-dimensional model of the apartment was created from building plans and photographs using Blender, a standalone 3D modelling software. Materials such as concrete, glass and wood were assigned to the corresponding walls, floors, windows and objects, and the resulting model was imported into Sionna.

In this annex, the digital twin comprises the three-dimensional geometric and material model of the apartment together with the Sionna ray-tracing configuration used to calculate indoor propagation.

Figure C.2 (a) shows the indoor model created in Blender. Figure C.2 (b) shows the corresponding scene after importation into Sionna, including the transmitter and receiver configuration.



(a) (b)

Figure C.2: Reference indoor environment: (a) three-dimensional model created in Blender; (b) corresponding scene imported into Sionna with the transmitter and receiver configuration

The digital twin was validated at 5,2 GHz using an omnidirectional reference transmitter. RSS measurements were performed in the real apartment at 44 receiver positions. In the digital twin, the corresponding omnidirectional transmitter model was placed at the same transmitter location, and the receiver coordinates were selected to match the 44 measurement positions.

The Sionna ray-tracing calculation included line-of-sight, reflection and refraction propagation components. Figure C.3 shows the resulting simulated RSS radio map. Figure C.4 compares the simulated and measured RSS values at the 44 receiver positions.

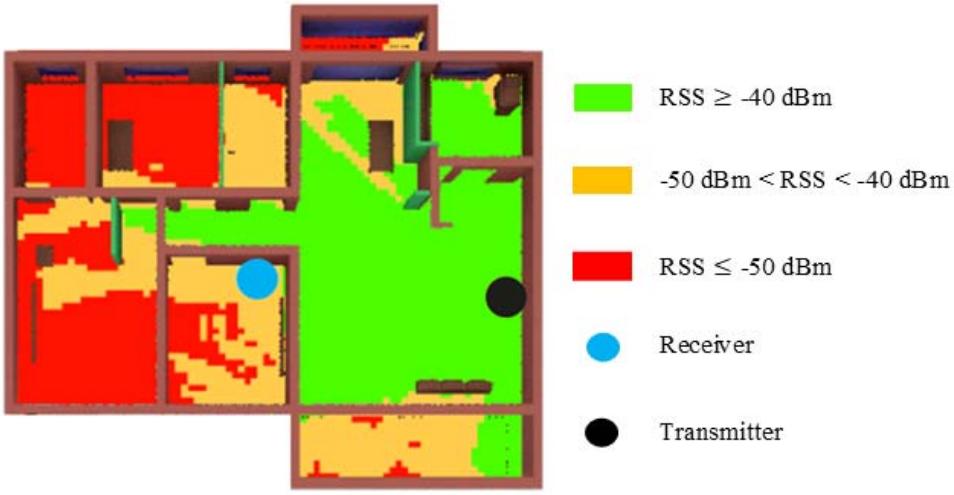


Figure C.3: RSS radio map of the omnidirectional antenna simulated in digital twin at 5,2 GHz

NOTE: The colour ranges used in Figure C.3 are visualization ranges for the digital-twin validation and are distinct from the receiver-adapted performance classes applied in clause C.5.

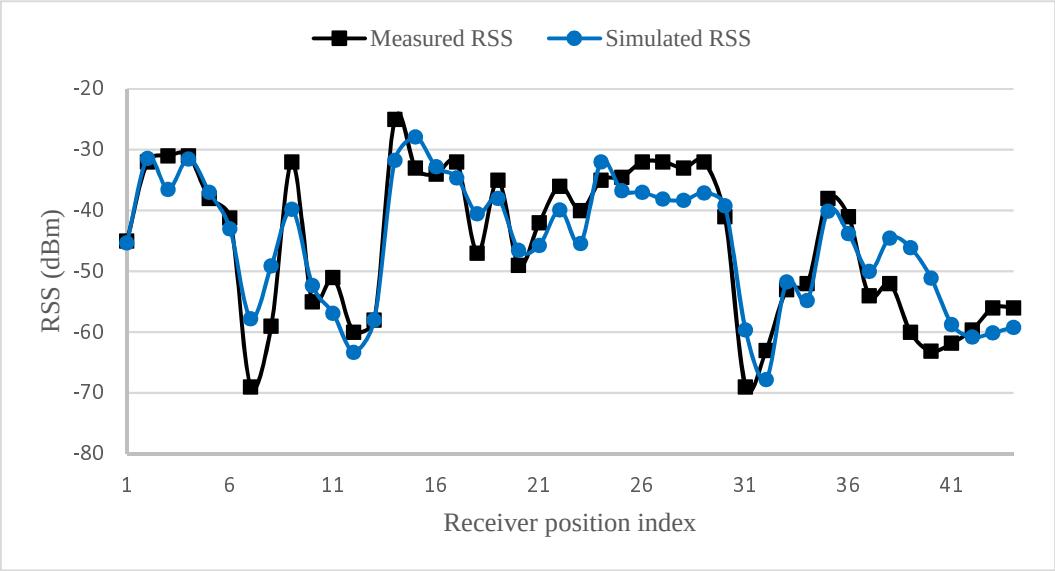


Figure C.4: Simulated and measured RSS at the 44 receiver positions used for digital-twin validation at 5,2 GHz

The mean absolute RSS difference over the 44 receiver positions is calculated as:

$$\Delta_{\mathrm{MAE}} = \frac{1}{44}\sum_{i=1}^{44}\left|RSS_{\mathrm{measured},i} - RSS_{\mathrm{simulated},i}\right| \quad \text{(C.1)}$$

For the data presented in Figure C.4, using equation (C.1), the resulting mean absolute RSS difference is 4,36 dB. This value represents the aggregate end-to-end discrepancy between the measured and simulated RSS values and can include contributions from measurement uncertainty, transmitter and receiver positioning, geometric representation of the environment, assigned material properties, antenna modelling and ray-tracing configuration. The individual contributions are not separately quantified in this example. This comparison supports the use of the indoor model and ray-tracing configuration for the hybrid measurement-simulation evaluation presented in the following clause. The same 44 receiver positions are retained for the comparison between the simulation-only and hybrid results and for computation of the performance criteria.

The measured RSS values used for the digital-twin validation span approximately -69 dBm to -25 dBm across the 44 receiver positions, while the corresponding simulated values span approximately -67,8 dBm to -27,9 dBm. The comparison therefore covers a broad range of received signal levels, but it does not include measured values below -70 dBm. The validation conclusion is therefore limited to the RSS range represented by the 44 measurement positions.

C.5 Hybrid Indoor Coverage Evaluation

C.5.1 Introduction

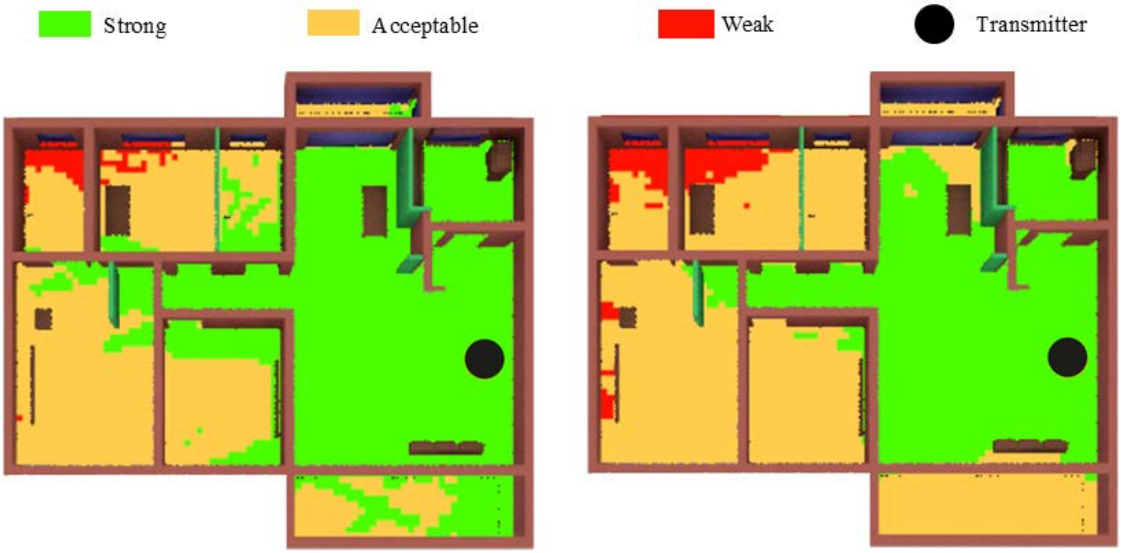
Two indoor propagation calculations were performed at 5,18 GHz using the validated residential digital twin and a configured transmitter power of 23 dBm. The same transmitter power was used in both cases. The simulation-only case used the CST-simulated complex far-field pattern of the selected SFU antenna. The hybrid measurement-simulation case used the complex far-field pattern calculated from the equivalent sources derived from the near-field measurements.

The same indoor environment, transmitter position, receiver positions and ray-tracing configuration were used in both cases. The antenna radiation pattern used as the transmitter input was the only modified element.

C.5.2 Coverage Maps

Figure C.5 compares the RSS coverage obtained for the two cases. Figure C.5 (a) shows the simulation-only result obtained using the CST-simulated far-field pattern. Figure C.5 (b) shows the hybrid result obtained using the far-field pattern calculated from the near-field measurements.

The broad spatial coverage trend is preserved in both cases. However, the hybrid result predicts a smaller strong-signal region and larger acceptable-signal regions, particularly in the parts of the apartment farther from the transmitter. These differences show the effect of using the measured radiation behaviour of the complete SFU instead of its numerical antenna model.



(a) (b)

**Figure C.5: Indoor RSS coverage at 5,18 GHz:
(a) simulation-only result using the CST-simulated far-field pattern;
(b) hybrid measurement-simulation result using the far-field pattern calculated from the near-field measurements**

The coverage maps apply the receiver-adapted RSS classes defined in Table C.4:

- Strong: $RSS \geq -52{,}15$ dBm
- Acceptable: $-72{,}15$ dBm $\leq RSS < -52{,}15$ dBm
- Weak: $RSS < -72{,}15$ dBm

Green, orange and red represent the strong, acceptable and weak classes, respectively. The same classes are used for the quantitative evaluation at the 44 receiver positions.

C.5.3 Evaluation at 44 Receiver Positions

For radio-map generation, Sionna evaluates the received power at each map cell using an ideal isotropic dual-polarized receiver. For the threshold adaptation applied in this annex, the receiver is therefore represented by $G = 1$, with no additional polarization- or impedance-mismatch loss, i.e. $\eta_{\mathrm{pol}} = 1$and $\eta_{\mathrm{m}} = 1$.

Applying equation (8) to the receiver configuration used in this example gives the following threshold correction:

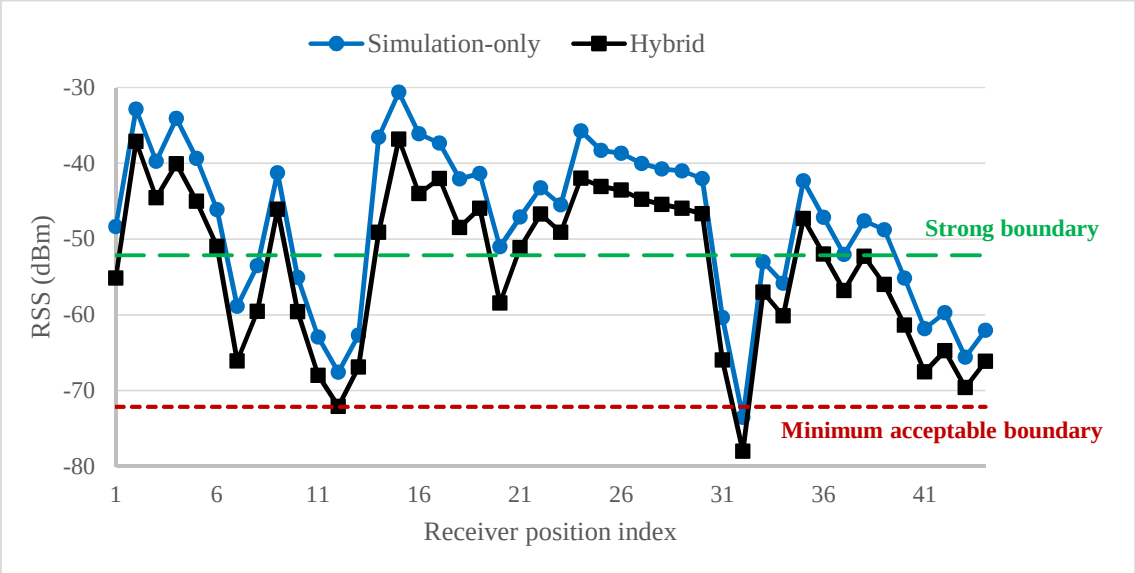
$$10log_{10}\left(\frac{1}{1{,}64}\right) = -2{,}15\ \mathrm{dB} \qquad (C.2)$$

The adapted criteria used for the evaluation are summarized in Table C.4.

Table C.4: Criteria applied to the Sionna receiver

Criterion	Applied Value
Strong boundary: T_{strong}	-52,15 dBm
Minimum acceptable boundary: T_{min}	-72,15 dBm
Average-RSS threshold: T_{avg}	-62,15 dBm
Required covered percentage: C_{req}	90 %

Figure C.6 compares the RSS values obtained at the same 44 receiver positions. The simulation-only RSS is higher than the hybrid RSS at every position, with a mean simulation-only-minus-hybrid difference of 5,33 dB. Nevertheless, the position-dependent variations follow a similar general trend in both cases.



NOTE: The receiver-adapted strong and minimum acceptable boundaries defined in Table C.4 are also shown.

Figure C.6: Simulation-only and hybrid RSS at the 44 receiver positions at 5,18 GHz

Compared with the simulation-only case, the hybrid case reduces the number of positions classified as strong from 29 to 24 and increases the number classified as acceptable from 14 to 19. One position is classified as weak in each case. The corresponding metrics are summarized in Table C.5.

Table C.5: Performance metrics at the 44 receiver positions

Case	C_{strong}	$C_{acceptable}$	C_{weak}	$C_{covered}$	RSS_{avg}	Result
Simulation-only	65,9 %	31,8 %	2,3 %	97,7 %	-40,4 dBm	Pass
Hybrid	54,5 %	43,2 %	2,3 %	97,7 %	-46,0 dBm	Pass

Both cases satisfy the reference residential criteria. The covered percentage is 97,7 % in both cases, exceeding the value of 90 %. The linear-power-averaged RSS values are -40,4 dBm for the simulation-only case and -46,0 dBm for the hybrid case, both above the applicable threshold of -62,15 dBm.

The hybrid result therefore satisfies the two performance criteria while predicting a lower average received power and a lower proportion of strong-signal positions than the simulation-only result.

C.6 Summary

The example in this annex combines a free-space comparison of the measurement-derived and simulation-derived radiation characteristics with validation of the indoor digital twin. In the final indoor evaluation, the simulation-only and hybrid cases show similar overall spatial and position-dependent RSS trends. The hybrid case nevertheless shifts some receiver positions from the strong to the acceptable class and yields a lower linear-power-averaged RSS, while the covered percentage remains the same in both cases. Both cases satisfy the spatial-coverage and average-RSS criteria defined in clause 5.4. The results therefore show that using the measurement-derived radiation pattern preserves the main coverage behaviour while revealing differences associated with the measured radiation characteristics of the complete SFU.

Annex D:
Change history

Date	Version	Information about changes
February 2026	V0.0.1	First populated Early Draft including methodology framework, test case definition, Annex A comparison of methods and Annex B digital twin framework.
May 2026	V0.0.2	Revised draft incorporating editors' comments, definition of performance criteria, antenna adaptation methodology, and structural improvements.
July 2026	V0.0.3	Stable Draft introducing spatial RSS range classification, combined coverage and average-RSS performance criteria, receiving-antenna gain, polarization and impedance-mismatch adaptation, and a validation and application annex.
August 2026	V0.0.4	Revised stable draft incorporating technical and editorial revisions, including clarification of RSS performance criteria and reference thresholds, expanded uncertainty considerations, equation-numbering corrections, and additional Annex C validation information.
August 2026	V0.0.5	Proposed final draft for working group approval addressing review comments on V0.0.3 and incorporating the corresponding technical, structural and editorial revisions.
August 2026	V0.0.6	Outcome: final draft from F5G#27; some minor editorial updates were made in the title, and a hanging paragraph was removed between C.5 and C.5.1.

History

Version	Date	Status
V1.1.1	September 2026	Publication